

Algebra MATH-310

Lecture 4

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Plan of the course

- 1 Integers: 1 lecture
- 2 Groups: 6 lectures
- 3 Rings and fields: 5 lectures
- 4 Review: 1 lecture

Today: Groups: lecture 3

- a Groups in cryptography: Elliptic curves and Lenstra's factorization algorithm
- b Non-abelian groups: the dihedral group
- c Normal subgroups and quotients
- d Examples of quotient groups

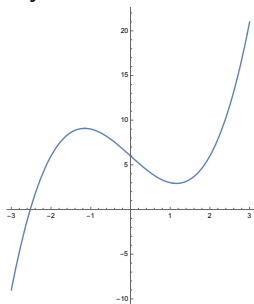
Elliptic curve group

Definition

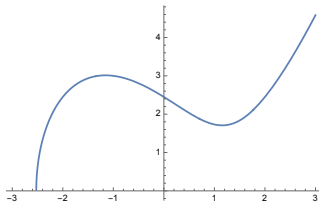
$$\mathbb{K} = \mathbb{R}$$

An elliptic curve is a subset of points in a plane $\mathbb{K}^2$ that satisfy the equation $y^2 = x^3 + ax + b$ where $a, b \in \mathbb{K}$. $(4a^3 + 27b^2 \neq 0)$

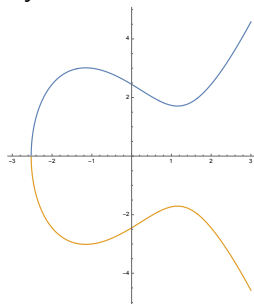
$$y = x^3 - 4x + 6$$



$$y = \sqrt{x^3 - 4x + 6}$$



$$y^2 = x^3 - 4x + 6$$

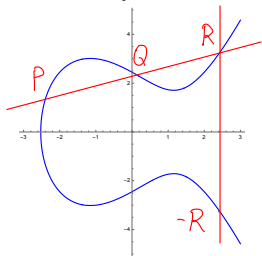


The set of $\mathbb{K}$ -rational points on an elliptic curve has a **group structure**

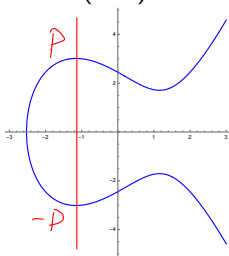
Set $P + Q = -R$ whenever three points P, Q, R are collinear points on the curve.

- 1 The neutral element is the point O "up at ∞ "
- 2 For any P the opposite point is the point symmetric to P with respect to the horizontal axis. Then $P + (-P) = O$.
- 3 For any P , we have $P + O$ (vertical line through P) intersects the curve in $-P$, so we have $P + O = -(-P) = P$.

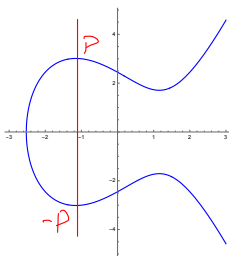
$$P + Q = -R$$



$$P + (-P) = O$$

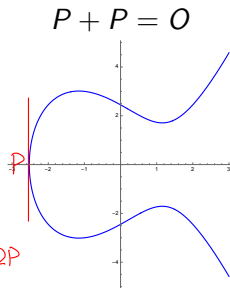
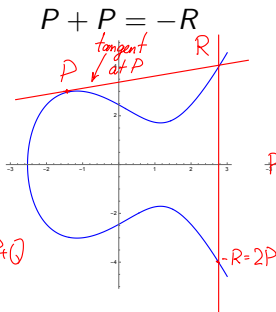
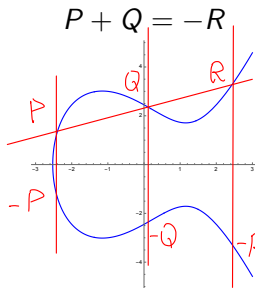


$$P + O = P$$



Elliptic curve group

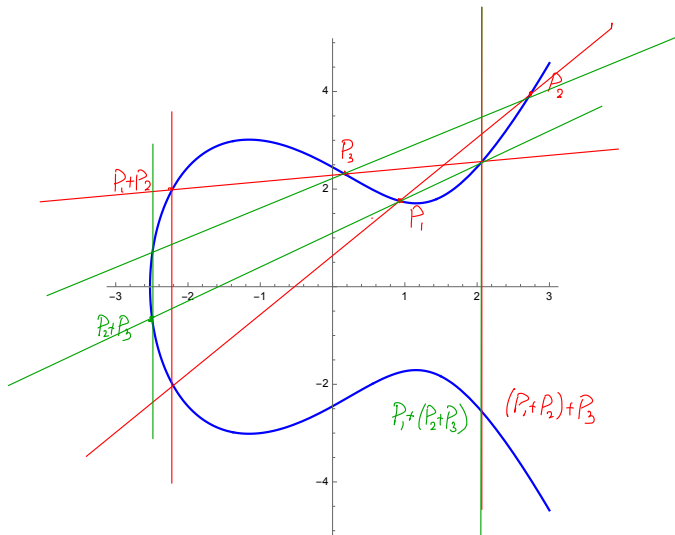
- 4 If P, Q, R are three intersection points of a line with the curve, then $P + Q = -R$, $Q + R = -P$ and $P + R = -Q$.
- 5 To find $P + P$, draw a tangent to the curve at P which intersects the curve at R . Then $P + P = -R$, and $P + R = -P$.
- 6 If P has y -coordinate zero, then $P + P = O$.



Elliptic curve group $(P_1 + P_2) + P_3 = P_1 + (P_2 + P_3)$

The addition is clearly commutative. It is also associative (harder to show).

Example:



Elliptic curve group: conclusions

- 1 Computing sums and multiples of points involves computing slopes of lines of the form $\frac{u}{v}$.
- 2 If we consider the curve over numbers $\mathbb{Z}/n\mathbb{Z}$, then the construction fails if and only if v is not invertible modulo n . This is exactly when $\gcd(v, n) > 1$. This is the idea of Lenstra's factorization algorithm (Hendrik Lenstra, 1987).



collaborated with
Arjen Lenstra
(EPFL)

Lenstra's factorization algorithm

Suppose you want to factorize a number $n \in \mathbb{N}$.

- 1 Pick up an elliptic curve $y^2 = x^3 + ax + b$ over $\mathbb{Z}/n\mathbb{Z}$ and a point P on it. *a, b integers* *with integer coord.*
- 2 Compute $2P, 3!P, 4!P$, etc until the computation fails.

$$3!P = 3 \cdot 2P = 2 \cdot 2P + 2P, \dots$$

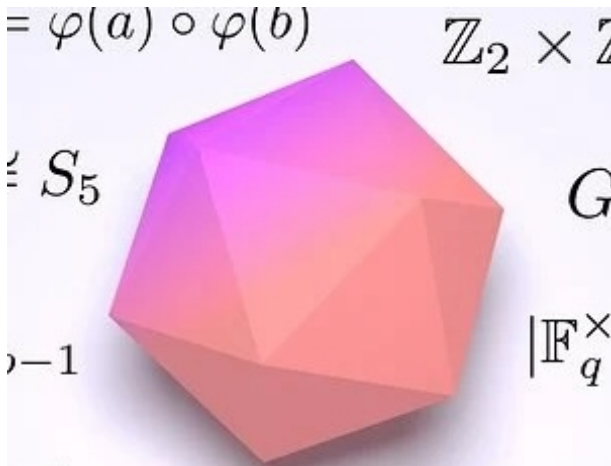
This involves computing slopes of lines $\frac{u}{v}$ modulo n , which makes sense if and only if $\gcd(v, n) = 1$ and v is invertible in $\mathbb{Z}/n\mathbb{Z}$.

- 3 If the computation fails, this implies $\gcd(v, n) > 1$ and you have found a nontrivial factor of n .
- 4 Otherwise restart with a different curve and point P .

the best factorization algorithm for n with < 50 digits

This method is especially efficient to find small factors of n .

Back to group theory!



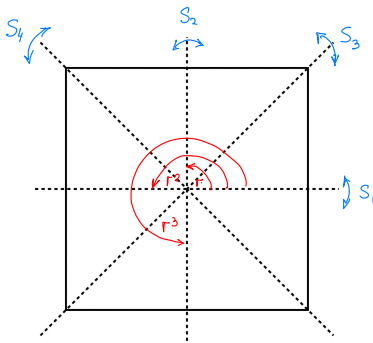
Dihedral group

Definition

D_n , $n \geq 3$, is the group of rigid symmetries of a flat regular n -gon.

$$D_n = \{1, r, \dots, r^{n-1}, s_1, s_2, \dots, s_n\}.$$

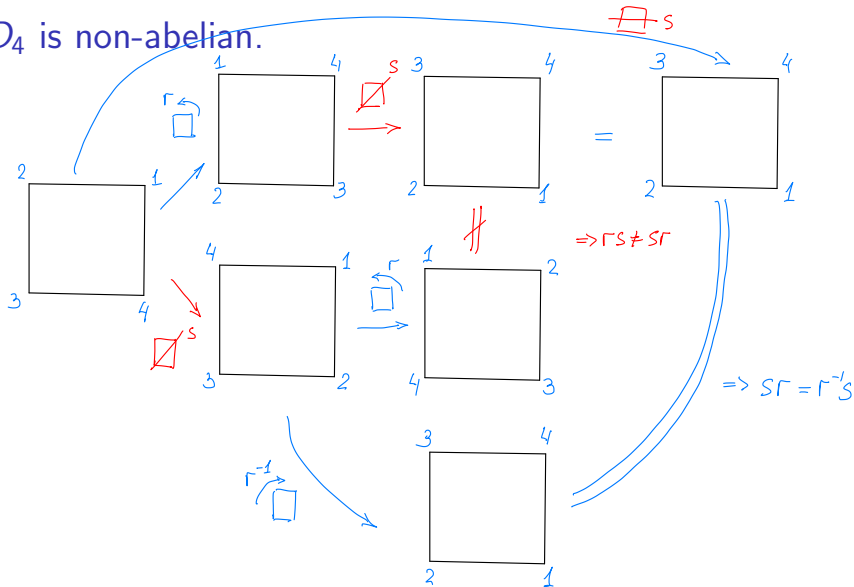
Example: $D_4 = \{1, r, r^2, r^3, s_1, s_2, s_3, s_4\}$



at least $2n$
elements

a group with respect to compositions

D_4 is non-abelian.



$\Rightarrow sr = r^{-1}s$ is a relation in D_4

Number of elements and relations in D_n :

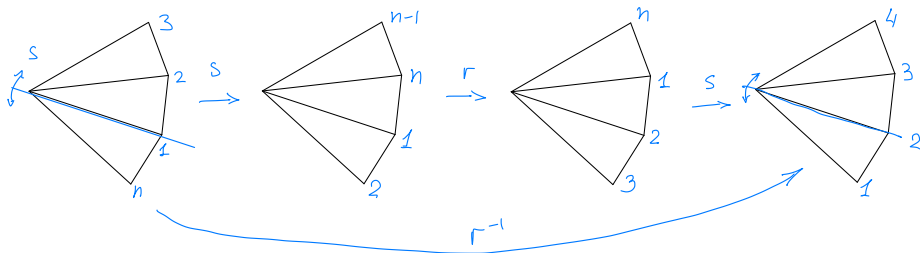
Vertex 1 $\rightarrow n$ possibilities

Vertex 2 $\rightarrow 2$ possibilities

But we have already found $2n$ elements

Relations in D_n

Let s be a reflection through a vertex, r a counterclockwise rotation by $\frac{2\pi}{n}$.



Conclusion: $srs = r^{-1}$. $\Leftrightarrow sr = r^{-1}s \Leftrightarrow sr sr = 1 \Leftrightarrow (sr)^2 = 1$

Presentation of D_n in generators and relations

Proposition

D_n admits a presentation in generators and relations:

$$D_n = \langle s, r \mid s^2 = 1, r^n = 1, srs = r^{-1} \rangle.$$

Complete list of elements: $\{ \underbrace{1, r, r^2, \dots, r^{n-1}}_{\text{rotations}}, \underbrace{s, sr, sr^2, \dots, sr^{n-1}}_{\text{reflections}} \}$ Exercise: $(sr^j)^2 = 1$
 $\forall j=0 \dots n-1$

Idea: $sr = r^{-1}s \Rightarrow$ Any product

$sr^3 sr^{-2} s \dots$ can be written in the form $r^b s^a$, $s^2 = 1$

$\Rightarrow$ get only $\{ r^i, sr^j \}_{\substack{i=0 \dots n-1 \\ j=0 \dots n-1}} \Rightarrow$ get $2n$ elements

Any additional relation would reduce # of elements in the group

What do we need it for?

Recall: A subgroup $H \subset G$ is **normal** in G if $ghg^{-1} \in H$ for any $g \in G$, $h \in H$. If G is abelian, then any subgroup $H \subset G$ is normal:
 $ghg^{-1} = gg^{-1}h = h \in H$.

If G is non-abelian, the situation is more interesting.

Examples of subgroups in D_n

$$D_n = \langle s, r \mid s^2 = 1, r^n = 1, srs = r^{-1} \rangle.$$

- ① Let $R = \langle r \rangle = \{1, r, \dots, r^{n-1}\} \subset D_n$ be the subgroup of rotations.

Then $R \trianglelefteq D_n$ is **normal** in D_n :

Need to check: $gr^k g^{-1} = r^j \quad \forall g \in D_n$; $r r^k r^{-1} = r^k \in R$

$$s r^k s^{-1} = s r^k s = \underbrace{(s r s)(s r s)(s r s)}_K \dots \underbrace{(s r s)}_{s r s = r^{-1}} = \underbrace{r^{-1} r^{-1} \dots r^{-1}}_K = r^{-k} \in R$$

Recall: If $H \subset G$ a subgroup, then the left coset $gH = \{gh, h \in H\}$.

Cosets in D_n with respect to R :

$$1R = \{1, r, \dots, r^{n-1}\} = r^2 R = \{r^2, r^3, \dots, 1, r\}$$

$$sR = \{s, sr, \dots, sr^{n-1}\} = sr^3 R = \{sr^3, sr^4, \dots, s, sr, sr^2\}$$

$$D_n = (1R) \cup (sR) \text{ disjoint union}$$

- ② Let $K = \langle s \rangle = \{1, s\} \subset D_n$. Then K is **not normal** in D_n :

$$s s s^{-1} = s \quad ; \quad (r s) r^{-1} = s r^{-1} r^{-1} = s r^{-2} \notin K.$$

$$s r s = r^{-1} \Rightarrow (r s) = s r^{-1}$$

Poll: normal subgroups in D_n $n = 12$

Consider the following subgroups in dihedral groups:

✓ (1). $\{1, r^3, r^6, r^9\} \subset D_{12}$

$$sr^3s = r^{-3} = r^9$$

✓ (2). $\{1, r^6\} \subset D_{12}$

$$sr^6s = r^{-6} = r^6$$

(3). $\{1, sr^6\} \subset D_{12}$

$$rsr^6r^{-1} = sr^{-1}r^6r^{-1} = sr^4 \notin \text{subgp}$$

✓ (4). $\{1, r^2, r^4, r^6, r^8, r^{10}\} \subset D_{12}$

$$sr^{2k}s = r^{-2k}$$

(5). $\{1, sr^4\} \subset D_{12}$.

$$rsr^4r^{-1} = sr^{-1}r^4r^{-1} = sr^{-2} \notin \text{subgp}$$

Poll: The following subgroups in the list are normal:

A: Only (1), (2) and (3).

B: Only (1), (2) and (4).

C: Only (2) and (4)

D: Only (1).

E: Only (1), (3) and (5).

Quotient groups: group of cosets with respect to a normal subgroup

Proposition

If $H \trianglelefteq G$ is a normal subgroup, then the set of left H -cosets in G naturally forms a group. Namely, define $(xH) \cdot (yH) = (xyH)$, then $(1H)$ is the neutral element and $(xH)^{-1} = (x^{-1}H)$. The group law on the set of left H -cosets G/H is well defined and gives rise to a group structure on G/H .

Need to check: the group law does not depend on the choice of representatives of cosets.

Suppose $x' \in xH$, $y' \in yH$ Need to show: $x'y' \in xyH$

$$x' = xh_1, \quad y' = yh_2 \Rightarrow x'y' = xh_1yh_2 = xy(\underbrace{y^{-1}h_1y}_{y^{-1}h_1y = h_3 \in H})h_2 = xyh_3h_2 \in xyH.$$

$\Rightarrow$ indeed, $x'y' \in xyH \Rightarrow$ the group is well defined. 

Example: Cosets of $R = \{1, r, \dots, r^{n-1}\}$ in D_n .

$$1R = \{1, r, \dots, r^{n-1}\}$$

$$sR = \{s, sr, \dots, sr^{n-1}\}$$

These two cosets form a group of two elements, called the **quotient group** D_n/R with elements $\{1R, sR\}$ and the group law defined by the proposition above.

Neutral element: the coset of 1.

$$(1R)(sR) = (sR)$$

$$(sR)(1R) = (sR)$$

$$(sR)(sR) = (1R)$$

$$(1R)(1R) = (1R)$$

$$\text{Let } g_1 = sr^i \in sR \quad g_2 = r^j \in 1R$$

$$g_1 g_2 = sr^i \cdot r^j = sr^{i+j} \in sR \Rightarrow (sR)(1R) = (sR)$$

Note that D_n/R is abelian.

$$g_2 g_1 = r^j sr^i = \underbrace{(sr^j s)}_{=r^{-j}} r^i = sr^{i-j} \in sR \Rightarrow (1R)(sR) = (sR)$$

Conclusion: we obtained the group $D_n/R \simeq C_2 = \langle t \mid t^2 = 1 \rangle = \{1, t\}$.

Summary of elements of group theory

- ① A subgroup $H \trianglelefteq G$ is **normal** if $ghg^{-1} \in H$ for any $g \in G$, $h \in H$.
- ② A **quotient group** G/H is a group of left cosets with respect to a normal subgroup $H \trianglelefteq G$. Multiplication: $(xH)(yH) = (xyH)$.
 $|G| = |G/H| \cdot |H| = [G : H] \cdot |H|$ by **Lagrange's theorem**.
- ③ If $\phi : G_1 \rightarrow G_2$ is a **group homomorphism**, then $\ker \phi \trianglelefteq G_1$ is normal in G_1 .
- ④ Any subgroup $H \subset G$ is normal in an abelian group G .
- ⑤ $D_n = \langle s, r \mid s^2 = 1, r^n = 1, srs = r^{-1} \rangle$, $n \geq 3$ is an example of a non-abelian group.
Then $R = \langle r \mid r^n = 1 \rangle \trianglelefteq D_n$ is normal and $D_n/R \simeq C_2$ is a cyclic group of 2 elements.
 $|D_n| = [D_n : R] \cdot |R| = 2 \cdot n = 2n$.